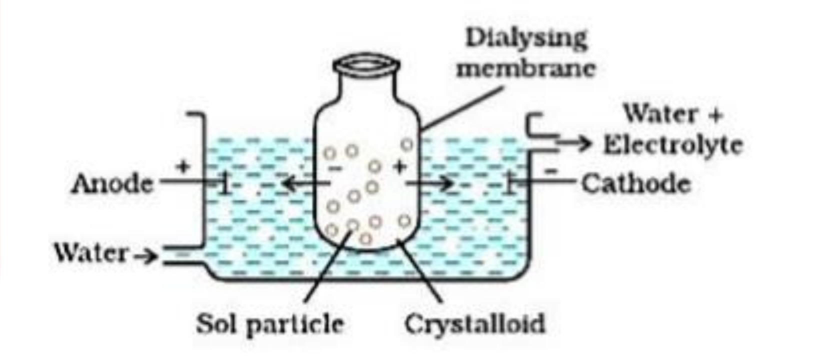
Rajasthan **Board Class 12**

Subject – Chemistry

Time: $3\frac{1}{4}$ *Hours* **M.M: 56**

Section – A

1. **Which type of semiconductor is formed when Arsenic is dopped with Germanium ?**
   Answer:
   IFAs is dopped with Ge, p-type semiconductor is formed
2. **Write definition of osmotic pressure.**
   Answer:
   The amount of external pressure required to be applied on conc. side to stop movement of solvent When two liquids of different conc. are separated by SPM, solvent flows from low conc. to high conc is known as osomotic pressure.
3. **Draw a labelled diagram of dialysis method, for purification of colloidal solutions.**
   Answer:
   Diagram of dialysis method, for purification of colloidal solutions is shown below:



4. **Write electronic configuration of Chromium (Z = 24).**
   Answer:
   $Cr(24) = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$
5. **Write name and symbol of one transurenic element.**
   Answer:
   Transurenic element formed by decomposition of uranium like Neptunium.
6. **Write general oxidation state of Lanthanoids.**
   Answer:
   General oxidation state of lanthanides is +3
7. **Write IUPAC name of the following complex compound $K_3\left[Fe(C_2O_4)_3\right]$.**
   Answer:

Potassium trioxalatoferrate (III)

8. Write chemical equation of Wurtz reaction.

Answer:

Two molecules of alkyl halides (same or different) react with sodium in the presence of dry ether to form alkanes.

$$RX + 2Na + RX \xrightarrow[ether]{pure\ \&\ \text{dry}} R - R + 2NaX$$

$$CH_2Cl + 2Na + ClCH_3 \xrightarrow[ether]{pure\ \&\ \text{dry}} CH_3CH_3 + 2NaCl$$

9. Write full name of DDT.

Answer:

DDT – Dichloro Diphenyl Trichloroethane

10. Write IUPAC name and chemical formula of Acetone.

Answer:

IUPAC name of Acetone: Propan-2-one

And the chemical formula of Acetone is $\mathrm{CH_3COCH_3}$.

11. Write the formula to determine 'weight average molecular weight' of polymers.

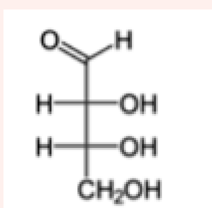
Answer:

$$\left(\bar{M}_n\right) = \frac{\sum n_1 m_1^2}{\sum n_1 m_1}$$

12. Write Fischer Projection Formula of erythrose sugar.

Answer:

Fischer Projection Formula of erythrose sugar $\left(\mathrm{C_4H_8O_4}\right)$.



13. Which chemical substance is found in musk, excreted by male musk deer?

Answer:

Pheromones

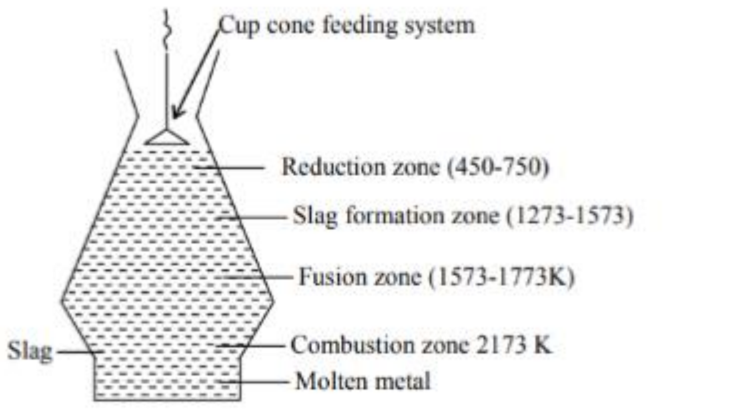
Section – B

14. (A) Draw a neat and labelled diagram of "blast furnace".

(B) What is the role of Silica in the copper metallurgy?

Answer:

(A) Neat labelled diagram of 'blast furnace' is shown below:



(B) SiO_2 is used as flux which remove FeO into slag $FeSiO_3$

$FeO + SiO_2 \rightarrow FeSiO_3$

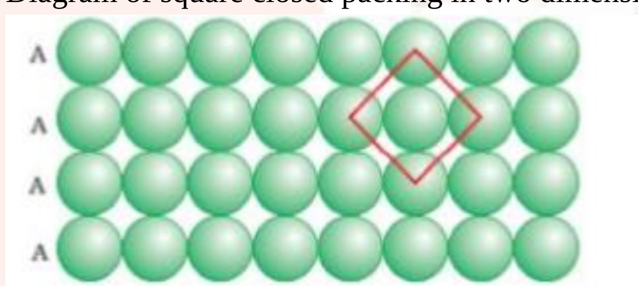
15. (A) Copper shows electrical conductance in solid as well as molten state whereas copper chloride shows electrical conductance only in molten state. Give reason.

(B) Draw a diagram of "two dimension square closed packing".

Answer:

(A) Cu metal has free e^- present which conduct electricity in both states. Where as in $CuCl_2$ ions conduct electricity which become free under molten state only.

(B) Diagram of square closed packing in two dimension is given below:



16. (A) Generally solubility of gases in liquids is decreases as increasing temperature. Give reason.

(B) How many gram of NaCl is required to make 200 mL aqueous solution of 5% (w/v) NaCl

Answer:

(A) Dissolution of gas in the liquid is an exothermic process with increase in temp. solubility decreases

(B) We know that,

$$\frac{W}{V}\% = \frac{mass\ \text{of solute} \times 100}{Volume}$$

$$\Rightarrow 5 = \frac{x}{200} \times 100$$

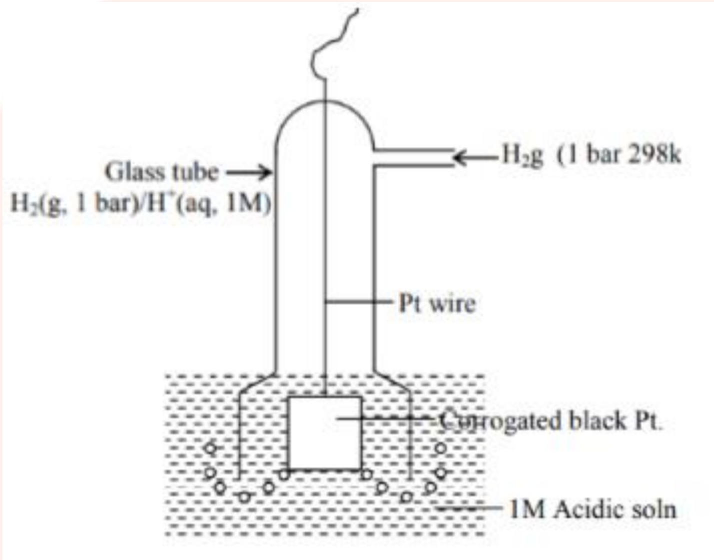
$$\Rightarrow x = 10g$$

17. (A) Draw a labelled diagram of 'Standard Hydrogen Electrode'.

(B) Fuel Cells are better than other cells. Give any two reasons.

Answer:

(A) Labelled diagram of 'Standard Hydrogen Electrode' is shown below:



(B) Fuel cells are better than other cells due to the following reasons,

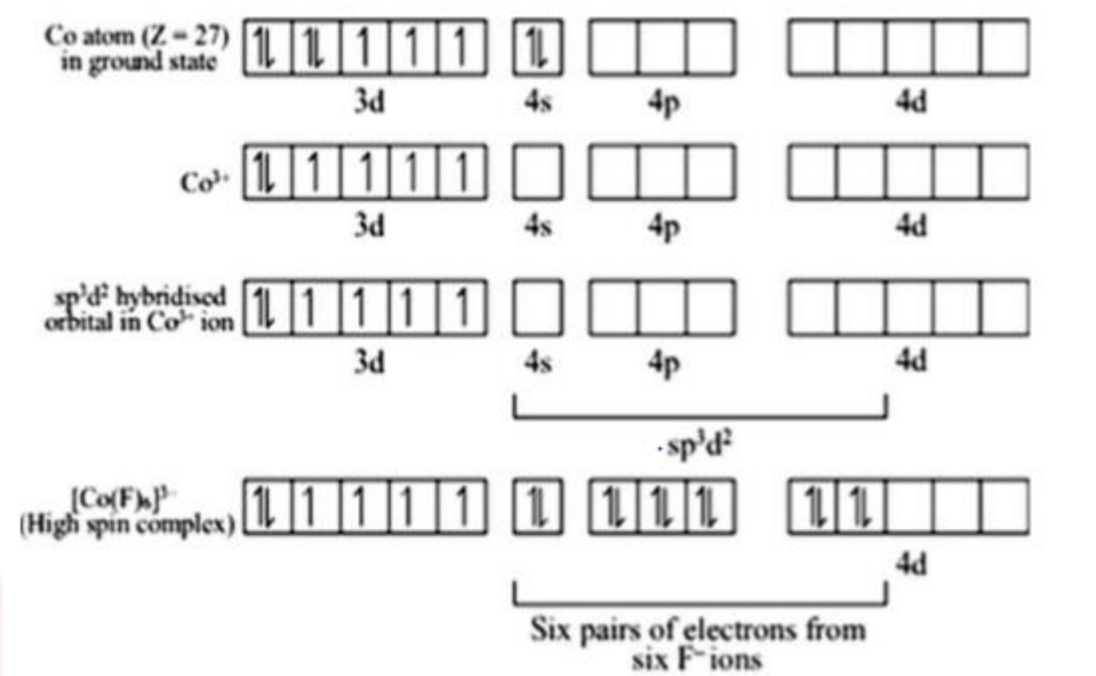
- Efficiency of fuel cell is high.
- No harmful emissions are released.

18. On the basis of valence bond theory explain the oxidation state, hybridisation, geometry and magnetic nature of metal in complex $[CoF_6]^{3-}$.

Answer:

The atomic number of Co is 27 and its valence shell electronic configuration is $3d^7 4s^2$.

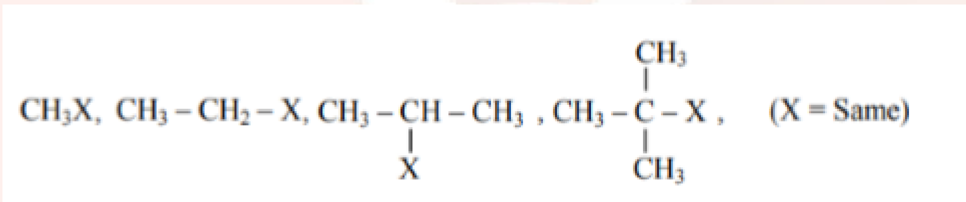
Co is in +3 oxidation state in the complex $[CoF_6]^3$.



Therefore,

$\left[CoF_6\right]^{3}$ Is sp^3d^2 hybridized and in octrahedral shaped.

19. (A) Arrange the following alkyl halides in ascending order of their reactivity towards SN_1 reaction .



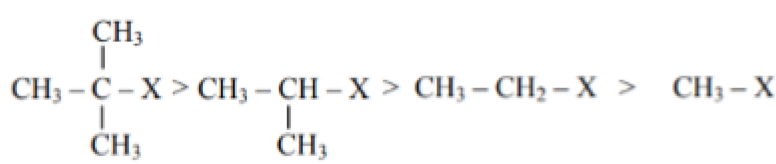
(B) Complete the following chemical reactions and write the products.

(i) $CH_3 - CH_2 - CL + KOH \longrightarrow$

(ii) $R - CH = CH_2 + HBr \xrightarrow{Peroxide}$

Answer:

(A) Reactivity of alkyl halides towards SN_1 nucleophilic substitution reaction is $3^0 > 2^0 > 1^0$ because in SN_1 nucleophilic reaction , the first and the slow step is the formation of a carbocation. Tertiary carbocation is more stable than a secondary carbocation which is more stable than a primary carbocation.



(B) (i) $CH_3-CH_2-Cl+KOH \rightarrow \underset{Ethane}{CH_2=CH_2}+KCl+H_2O$

(ii) $R-CH=CH_2+HBr \xrightarrow{Peroxide} \underset{Alkyl\ \text{bromide}}{R-CH_2-CH_2-Br}$

20. A solution of copper sulphate electrolysed for 20 minute with a current of 1.5 Ampere. Calculate the mass of copper deposited at the cathode. (F = 96500 C)

Answer:

The cathodic reaction is $\text{Cu}^{2+}+2e^- \rightarrow Cu$.

Therefore,

$2\times96500C$ Are required to deposit 63.5g Cu at cathode.

$\Rightarrow$1800 will deposit Cu=$\dfrac{63.5\times1800}{2\times9600}=0.592g$

21. (A) Arrange the following compounds in the descending order of their reactivity towards nucleophilic substitution reaction.

$CH_3CHO, HCHO, CH_3COCH_3$

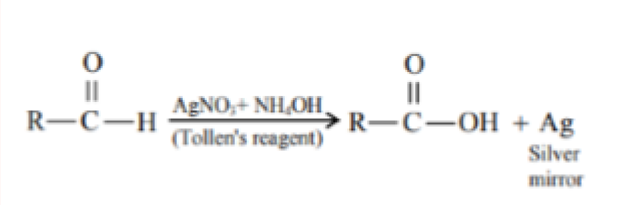
(B) Alkanoic acid have highest boiling points. Explain.

Answer:

(A) Electron-donating groups decrease the strengths of acids, while electron-withdrawing groups increase the strengths of acids. As methoxy group is an electron-donating group, 4methoxybenzoic acid is a weaker acid than benzoic acid.

4-Methoxybenzoic acid < Benzoic acid < 4-Nitrobenzoic acid

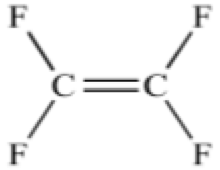
(B) Due to reducing nature aldehydes gives Tollen's test while ketones not



22. 22. (A) Write the name of monomer unit of polymer used in non-stick surface coated utensils. (B) Give an example of each of Homopolymer and copolymer.

Answer:

(A) Tetraflouroethylene



(B) Homo polymer: Poly ethylene or PVC

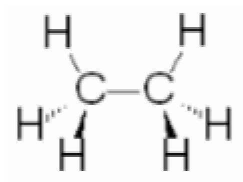
Co-polymer: Poly ester or Terylene or Nylon-66

23. 23 (A) **Which conformer has higher energy in the Sawhorse projection formula of ethane and why?**

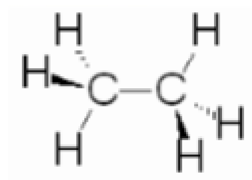
(B) Racemic mixture are optically inactive. Give reason.

Answer:

(A) Eclipsed conformer has higher energy than the staggered conformer due to torsional strain,



Eclipsed



Staggered

(B) Racemic mixture is in active due to the fact that in equimolecular mixture of enantiomeric pairs, the rotation caused by the molecules of one enantiomer is cancelled by the rotation caused by the molecules of other enantiomer.

24. **(A) Aspirin should not be taken in empty stomach, why?**

(B) Write any two differences between dyes and pigments.

Answer:

(A) Without eating anything aspirin get some stomach irritation, followed by a stomach ache. This happens because there is low acid level in the stomach which is maintained by the intake of food to dissolve the aspirin.

(B) Differences between dyes and pigments:

Most of dyes are organic. It has direct affinity to textile material. While, Most of pigments are inorganic. They have no direct affinity to textile material.

Section – C

25. **Read the given paragraph and write answers of the following questions. When any solid substance is kept in contact with liquid or gas, then liquid or gas are more adsorbed on the surface of solid rather than bulk. The process is known as adsorption. It is different from absorption. Many gaseous reaction occurs in the presence of solid catalyst. Catalyst is a chemical substance which change the rate of reaction without undergoing itself change. This phenomenon is known as catalysis.**

(A) Write any two differences between Absorption and Adsorption.

(B) Write any chemical equation of heterogeneous catalysis.

(C) Write the name of Zeolite catalyst used to convert alcohol to petrol.

Answer:

(A) Differences between Absorption and Adsorption are as follows:

Absorption	**Adsorption**

Assimilation of molecular species throughout the bulk of the solid or liquid is termed as	Accumulation of the molecular species at the surface rather than in the bulk of the solid or
absorption.	liquid is termed as adsorption.
It is a bulk phenomenon.	It is a surface phenomenon.
It is an endothermic process	It is an exothermic process.
It occurs at a uniform rate.	It steadily increases and reaches equilibrium.

(B) Oxidation of sulphur dioxide to form sulphur trioxide. In this reaction, Pt acts as a catalyst.

$$2SO_2 \xrightarrow{Pt} 2SO_3$$

(C) ZSM – 5 zeolite is useful for converting alcohol to petrol.

26. Read the given paragraph and write answers of the following questions. Protein is very essential for the growth, development and maintenance of living systems. Proteins are natural polymer of σ-amino acids. A definite sequence of amino acids form a specific protein. Two or more than two amino acids bind to give peptide bond. Proteins are polypeptide, which loses its biological activity by physical or chemical changes.

(A) Explain essential and non-essential amino acid with example.

(B) Explain denaturation of protein.

Answer:

(A) Essential amino acids are required by the human body, but they cannot be synthesised in the body. And non-essential amino acids are also required by the human body, but they can be synthesized in the body

(B) Denaturation is a process in which proteins lose the quaternary structure, tertiary structure and secondary structure which is present in their native state

27. (A) Complete the following equations and identify A and B.

$$R-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{Br_2 + KOH} [A] \xrightarrow{CHCl_3 + KOH} [B]$$

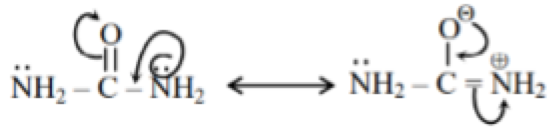
(B) Draw the resonating structure of Urea

Answer:

(A) Complete reaction is:

$$R-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2 \xrightarrow{Br_2+KOH} \underset{[A]}{R-\ddot{N}H_2} \xrightarrow{CHCl_3\,+\,KOH} \underset{[B]}{R-NC}$$

(B) The resonating structure of urea is:



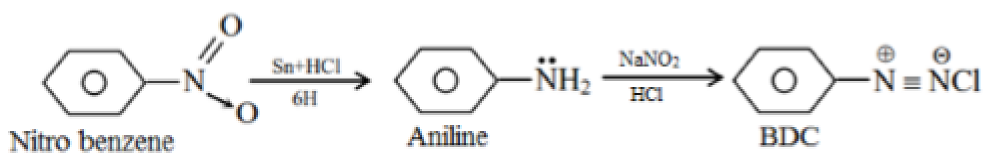
OR

27. A) Complete the following equations and identify A and B.

$$C_6H_5NO_2 \xrightarrow[6H]{Sn/HCl} [A] \xrightarrow{NaNO_2+HCl} [B]$$

(B) Draw the resonating structures of Aniline.

Answer:

(A) 

(B) Resonating structure of aniline is: